

Periodic and Fourier Structure of the Irrational Furstenberg Skew Shift

Route-A structural certificate C169

Abstract

For every irrational α , we solve the affine iterates of $T_\alpha(x, y) = (x + \alpha, y + x)$ on the two-torus. No positive iterate has a fixed point, so the Artin–Mazur zeta is one. On Haar L^2 , the Koopman operator acts explicitly on Fourier modes and admits an involutive reversal. This establishes a sharp contrast between natural operator structure and an empty periodic-orbit ledger.

Keywords: Furstenberg skew shift; irrational rotation; Koopman operator; Fourier decomposition; bilateral shift; time reversal.

中文摘要

本文对任意无理参数的二维Furstenberg斜移给出完整的迭代公式。任意正迭代均无不动点，因此Artin–Mazur动力 ζ 函数恒等于一。在Haar平方可积空间上，Koopman算子的傅里叶作用可以严格分解为纯点部分与双边移位部分；同时存在显式的对合时间反演。这说明自然算子结构不能弥补空的周期轨道数据。
关键词：Furstenberg斜移；无理旋转；Koopman算子；傅里叶分解；双边移位；时间反演。

1 Iterates and periodic points

Induction gives

$$T_\alpha^n(x, y) = \left(x + n\alpha, y + nx + \binom{n}{2}\alpha \right) \pmod{1}.$$

A fixed point would imply $n\alpha = 0 \pmod{1}$, impossible for irrational α . Thus $\# \text{Fix}(T_\alpha^n) = 0$ for every $n \geq 1$ and

$$\zeta_{AM}(z) = \exp \left(\sum_{n \geq 1} \# \text{Fix}(T_\alpha^n) z^n / n \right) = 1.$$

2 Fourier action and reversal

For $e_{m,k} = e^{2\pi i(mx+ky)}$ and $Uf = f \circ T_\alpha$,

$$Ue_{m,k} = e^{2\pi im\alpha} e_{m+k,k}.$$

The $k = 0$ sector is pure point. For $k \neq 0$, residue classes of m modulo $|k|$ give $|k|$ invariant weighted bilateral shifts. Their unimodular weights are removed recursively by diagonal unitaries, so every chain is unitarily equivalent to the bilateral shift. Summing over all $k \neq 0$ gives Lebesgue spectrum of countably infinite multiplicity. Moreover $R(x, y) = (\alpha - x, y)$ obeys $R^2 = I$ and $RT_\alpha R = T_\alpha^{-1}$, so complex conjugation after R is an antiunitary reversal.

3 Operator-ideal boundary

The Fourier basis is carried bijectively to another orthonormal basis. Thus U is not compact. Every singular value of a unitary equals one, so U belongs to no finite Schatten class. For $z \neq 0$, zU is not trace class; therefore the ordinary Fredholm determinant $\det(I - zU)$ is unavailable.

4 Boundary

The exact source zeta has no primitive cycles. No target divisor, arithmetic local factor, Hilbert–Pólya operator, or Route-B conclusion is claimed. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.